COMPLETE SPECIFICATION

(Under Section 10 and Rule 13 of the Patents Act, 1970)

TITLE OF THE INVENTION

Explainable Multimodal Financial Reasoning with Affective Signal Fusion for Market Decision Intelligence

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This Complete Specification is filed pursuant to Section 10 of the Patents Act, 1970, and constitutes the formal patent specification for the invention titled *Explainable Multimodal Financial Reasoning with Affective Signal Fusion for Market Decision Intelligence*.

1. ABSTRACT

Financial markets are characterized by rapidly evolving conditions, elevated volatility, and strong behavioral influences emanating from communications by government policymakers, monetary authority officials, corporate executives, institutional investors, and social media participants. Existing financial prediction systems based on quantitative indicator analysis fail to capture these affective and communicative market signals and further suffer from a fundamental lack of interpretability that constrains their utility for regulated institutional deployment. The present invention discloses an Explainable Multimodal Affective Financial Reasoning System (EMAFRS) integrating quantitative financial time-series indicators and algorithmic signals with affective feature vectors extracted from market influencer communications through a role-stratified natural language processing and emotion detection pipeline. The system employs a volatility-aware adaptive fusion engine that dynamically modulates the relative contribution of quantitative and affective signals as a function of a discrete market regime classification derived from hidden Markov model inference over realized volatility, market entropy, and Hurst Exponent measurements. A multimodal transformer reasoning network generates calibrated market direction predictions and risk estimates under a joint volatility-aware optimization objective incorporating a regime-dependent risk penalty. An explainability module employing Integrated Gradients attribution, transformer attention weight visualization, and template-driven natural language generation produces structured human-interpretable explanations attributing each prediction to specific financial indicators, affective signals, influencer role categories, and market regime characteristics. The invention addresses the critical unmet need for financially grounded, behaviorally enriched, and interpretably transparent decision intelligence tools capable of operating reliably across diverse and rapidly changing market regimes.

2. FIELD OF THE INVENTION

The present invention pertains to the technical fields of artificial intelligence-driven financial analytics, affective computing, high-frequency market microstructure analysis, and explainable decision support systems.

The present invention is a **computer-implemented technical platform and market surveillance system** — not a mathematical method, algorithm, or computer program *per se*. The invention comprises distinct hardware and software modules operating

in combination to perform real-time market monitoring, manipulation detection, and investor protection. Each functional module processes physical market data streams, produces structured output signals, and communicates alerts to users through defined interfaces. The technical contribution of the invention resides in the *architecture and interaction* of these modules as a unified system, which produces technical effects — early manipulation detection, execution-aware risk alerts, and retail investor protection — that are not produced by the individual components in isolation.

More particularly, the present invention discloses a novel **AI Market Integrity Framework (AMIF)** hereinafter also referred to as the *Explainable Multimodal Affective Financial Reasoning System* (EMAFRS), which is implemented as a market surveillance platform integrating structured quantitative financial indicators — including microsecond-scale order book dynamics, bid-ask imbalance metrics, liquidity withdrawal signals, and alpha persistence measurements — with affective and sentiment signals derived from market influencers including government policymakers, central bank officials, corporate executives, institutional investors, and social media participants, to generate transparent, interpretable, and volatility-aware financial market surveillance and decision intelligence.

The invention further incorporates techniques from Explainable Artificial Intelligence (XAI) and high-frequency trading microstructure theory to ensure that predictions, classifications, manipulation risk assessments, and execution-aware alerts produced by the **technical platform** are accompanied by human-interpretable counterfactual reasoning traces. This enables practitioners, regulators, institutional risk officers, and retail investors to audit the basis upon which market manipulation alerts are generated and to act upon those alerts in the context of their order execution strategies. The mathematical formulations described in this specification represent the computational operations performed by specific hardware and software modules within the platform, and are not claimed as mathematical methods in the abstract.

3. BACKGROUND OF THE INVENTION

3.1 Overview of Financial Market Complexity and the Manipulation Problem

Financial markets represent one of the most complex, high-dimensional, and dynamically evolving systems encountered in applied computing and data science. Market prices reflect not only fundamental economic valuation metrics but also collective

investor sentiment, macroeconomic policy expectations, geopolitical developments, and the communicative behavior of influential individuals whose pronouncements can shift market equilibrium within minutes.

A particularly acute and underaddressed challenge in contemporary financial markets is the detection of coordinated market manipulation, including pump-and-dump schemes in which artificial emotional hype is deliberately generated across social media platforms and broadcast communication channels to inflate asset prices before manipulating actors liquidate positions at the expense of retail investors. Such schemes operate at the intersection of behavioral, linguistic, and quantitative market signals and therefore cannot be detected by any single-modality monitoring system.

3.2 Technical Problem That Existing Systems Cannot Solve

The technical problem addressed by the present invention may be precisely stated as follows. Current financial monitoring and prediction systems suffer from four fundamental and inter-related deficiencies that collectively render them incapable of detecting emotion-driven coordinated market manipulation in real time and communicating actionable, interpretable warnings to retail investors.

First, existing systems rely exclusively on price and volume data, ignoring the emotional and communicative signals generated by public actors including policymakers, corporate executives, and social media influencers. Because pump-and-dump schemes are initiated and sustained through coordinated emotional hype rather than fundamental price action, systems that monitor only numerical market data are structurally incapable of detecting the manipulation at its origin. The technical consequence is that existing systems can at most detect the price anomaly after the manipulation has already harmed retail investors, rather than identifying the manipulative hype pattern during its formation.

Second, existing sentiment analysis systems that do process textual or audio signals operate on a single modality independently and in isolation from market regime context. Simple bag-of-words or unimodal language model classifiers assign static importance weights to sentiment signals regardless of whether the market is operating in a normal, volatile, or manipulation-risk regime. This static weighting means that the same sentiment signal may be appropriately treated as noise during normal market conditions but is critically important during a suspected manipulation event. Existing systems possess no mechanism to elevate the salience of emotional hype

signals specifically during suspicious market regimes.

Third, no existing system combines emotional hype detection, order book imbalance monitoring, abnormal volume detection, and market regime classification into a single coupled manipulation risk model. Conventional approaches treat these as independent signals or apply them in sequential pipelines without feedback between components. The technical consequence is that the individual components each produce high false-positive rates in isolation, and the signals that jointly indicate pump-and-dump activity are never evaluated in combination within a unified risk quantification model.

Fourth, existing monitoring systems produce opaque alerts or raw probability scores that are incomprehensible to non-expert retail investors, who represent the population most harmed by pump-and-dump manipulation. No existing system generates counterfactual explanations that communicate to a retail investor precisely which observable signals triggered a manipulation alert and what alternative market conditions would have been necessary for no alert to have been generated.

3.3 Why Existing Methods Cannot Solve the Identified Problem

The foregoing technical deficiencies are not merely implementation limitations that could be resolved by straightforward application of existing methods. They reflect fundamental architectural constraints of prior art systems.

Statistical anomaly detection methods, including z-score based volume alerts and price deviation monitors, operate on individual numerical signals in isolation and cannot integrate cross-modal information. They generate manipulation alerts only after price distortion is observable, which is too late to protect retail investors.

Unimodal sentiment analysis systems, including transformer-based financial sentiment classifiers applied to news text, process a single communication modality and cannot correlate emotional signals with simultaneous order book dynamics or regime state. They further assign fixed credibility to all sources regardless of historical market impact, failing to distinguish coordinated influencer hype from organic market commentary.

Multimodal machine learning systems that have been proposed in the academic literature for general financial forecasting tasks assign static fusion weights to modalities determined during training and not updated as a function of current market regime.

Such systems cannot shift attention toward manipulation-relevant signals during suspicious market conditions. The technical mechanism required to achieve this regime-conditioned reweighting, namely the Regime-Adaptive Multimodal Attention formulation described in the present disclosure, does not exist in any prior art system known to the inventors.

Existing explainability tools for financial models, including post-hoc feature attribution methods, produce technical attribution scores over model features that are meaningful only to quantitative analysts and are inaccessible to the retail investor population that requires protection from market manipulation. No prior art system generates counterfactual natural language explanations specifically designed for non-expert comprehension in the context of manipulation detection.

3.4 Statement of Technical Gap and Prior Art Distinction

The technical gap addressed by the present invention is precisely the absence of a unified, closed-loop **technical platform** that jointly accomplishes the following four technical functions, none of which is achievable in isolation by existing methods and the combination of which produces a qualitatively superior technical outcome that could not have been predicted from knowledge of the individual components: continuous multimodal observation of market and communication signals; emotion-regime coupled manipulation risk quantification; regime-adaptive signal fusion that elevates manipulation-relevant signals during suspicious market states; and counterfactual citizen-accessible explanation of triggered manipulation alerts.

Prior art systems in sentiment-based trading, multimodal financial forecasting, explainable financial AI, and pump-and-dump detection each address one or at most two of these functions. Existing sentiment-based trading systems process text signals in isolation from market microstructure. Existing multimodal forecasting systems assign static modality weights independent of market regime. Existing explainable financial AI systems produce technical attribution scores inaccessible to retail investors. Existing pump-and-dump detection systems monitor price and volume anomalies without incorporating affective signals from influencer communications. No prior art system integrates all four technical capabilities within a **single closed-loop platform**, and no prior art teaches that regime-adaptive attention weighting, microstructure-affective risk coupling, and counterfactual citizen alerts must interact architecturally to achieve early manipulation detection before price distortion

becomes observable. The novelty of the present invention therefore resides in this specific architectural combination and the measurable technical effect it produces, not in any individual algorithm or mathematical formulation taken in isolation.

3.5 Industrial Applicability

The present invention is industrially applicable as a deployable technical platform for: securities market surveillance operated by financial regulators and exchange operators; real-time retail investor protection services delivered through financial applications; institutional trading risk management systems; algorithmic order management systems requiring execution-aware manipulation alerts; and compliance monitoring tools for financial institutions subject to market integrity regulations. The system processes live market data feeds and produces actionable outputs consumed by human users and automated systems, satisfying the industrial applicability requirement under the Patents Act, 1970.

4. SUMMARY OF THE INVENTION

The present invention provides a computer-implemented AI Market Integrity Framework (AMIF) comprising a unified, closed-loop, multimodal market surveillance and citizen protection system capable of detecting pump-and-dump manipulation patterns and suspicious market regimes through the coordinated interaction of three technically distinct and mutually reinforcing innovations, described herein as the First, Second, and Third Inventive Mechanisms.

First Inventive Mechanism — Regime-Adaptive Multimodal Attention (RAMA). Existing multimodal systems assign static fusion weights to input modalities determined during training and fixed during inference. The present invention introduces a regime-adaptive attention mechanism in which the fusion weight assigned to each input signal is a dynamic function of the currently detected market regime. Formally, the unified market state representation is:

$$M_t = \sum_i \alpha_i(R_t) X_i, \quad \alpha_i(R_t) = \frac{e^{(I_i - \lambda L_i + \gamma R_t)}}{\sum_j e^{(I_j - \lambda L_j + \gamma R_t)}}$$

where I_i is the predictive information value of signal i, L_i is its computational latency cost, R_t is the detected market regime index, and γ is a regime sensitivity parameter. This formulation causes the system to automatically prioritize manipulation-relevant

signals — including emotional hype indicators and order book imbalance — when the detected regime suggests elevated manipulation risk, and to deprioritize them during normal market conditions. This regime-conditioned dynamic reweighting is technically distinct from all prior art static fusion architectures and produces a qualitatively superior detection capability that could not be achieved by any fixed-weight multimodal combination.

Second Inventive Mechanism — Emotion-Regime Coupled Pump-and-Dump Detection. Conventional manipulation detection systems evaluate price anomalies or volume anomalies in isolation. The present invention introduces a coupled three-signal manipulation risk model that simultaneously evaluates emotional hype intensity, order book structural imbalance, and abnormal trading volume within a unified risk quantification formula:

$$\text{Risk}_t = w_1 S_t + w_2 \,\text{OBI}_t + w_3 \,\text{Vol}_{\text{anomaly}}$$

where S_t is the emotion-weighted hype score derived from multimodal influencer communications, OBI_t is the order book imbalance metric, and $\text{Vol}_{\text{anomaly}}$ is the abnormal trading volume indicator. When $\text{Risk}_t > \tau$, the system flags possible pump-and-dump activity. The technical novelty of this mechanism lies in the coupling of the emotional signal S_t — which is derived from cross-modal affective analysis of influencer communications — with structural market microstructure signals within a single risk scoring function. This coupling enables detection of the early stages of coordinated hype campaigns before price distortion becomes observable, which cannot be achieved by any prior art single-modality manipulation detection system.

Third Inventive Mechanism — Counterfactual Citizen Alert Explainability. Existing monitoring systems produce either raw probability scores or technical feature attributions that are inaccessible to retail investors without quantitative expertise. The present invention generates counterfactual explanations by comparing the system decision under the actual observed signal values against the system decision under hypothetical alternative signal values, expressed as: $\text{Decision}(M_t, S_t, R_t)$ versus $\text{Decision}(M_t, S_t+\Delta, R_t)$. This comparison directly identifies which signals caused the alert and generates natural language explanations of the form: "Unusual sentiment spike combined with abnormal trading volume detected. Possible pump-and-dump activity." Such explanations are communicated to retail investors asynchronously, protecting citizens who are not actively monitoring markets at the time the suspicious

activity occurs. This counterfactual citizen-accessible explanation mechanism is technically distinct from post-hoc feature attribution methods in the prior art, which produce technical scores rather than actionable natural language alerts.

Non-Obvious Closed-Loop Interaction. The essential inventive contribution of the present disclosure resides not merely in the individual mechanisms described above but in their specific architectural interaction as a closed-loop feedback system. The regime classification produced by the manipulation-aware regime detection engine directly modulates the attention weights of the RAMA fusion mechanism, which in turn produces the multimodal market state representation used as input to the manipulation risk model, whose output activates the counterfactual explanation engine to generate citizen alerts. This four-stage feedback loop — in which hype detection, regime classification, manipulation risk scoring, and citizen alert generation mutually inform and reinforce one another — produces a system-level capability that no combination of the individual prior art components could achieve and that a person skilled in the art would not have arrived at through ordinary design choices.

Compliance with Section 3(k) of the Patents Act, 1970. The present invention is a **technical system and platform**, not a mathematical method, business method, algorithm, or computer program *per se*. The mathematical formulations described in this specification — including the RAMA attention equation, the manipulation risk scoring formula, and the counterfactual comparison procedure — are implemented as specific functional modules within a deployed market surveillance platform that ingests physical market data, processes multimodal communication streams, and delivers actionable alerts to human users and automated execution systems. These formulations are claimed only as they are *implemented in and integrated within the technical architecture* of the AMIF platform, and are not claimed as abstract mathematical relationships. The system produces concrete technical effects — early pump-and-dump detection, reduced false alert rates, execution-aware microstructure alerts, and retail investor protection — that are measured outcomes of the deployed platform, not properties of the mathematical models in the abstract. Accordingly, the present invention falls outside the exclusion of Section 3(k) and is patentable subject matter under Indian patent law.

5. ADVANTAGES OF THE INVENTION

The present invention provides several technical and practical advantages over existing financial monitoring and prediction systems.

1. Early Detection of Market Manipulation. The system integrates multimodal affective signals derived from influencer communications with quantitative financial indicators and market microstructure signals including Order Book Imbalance, Order Flow Imbalance, and Liquidity Shock Indicators. This enables the early identification of coordinated hype-driven pump-and-dump schemes before significant price distortion occurs, thereby improving the timeliness of manipulation detection relative to all prior art single-modality or dual-signal systems.

2. Improved Interpretability and Transparency. By incorporating explainable artificial intelligence mechanisms, including Integrated Gradients feature attribution and counterfactual explanation generation, the system produces human-interpretable reasoning traces for each decision. This allows analysts, regulators, and investors to clearly understand the factors contributing to generated alerts and predictions, satisfying emerging regulatory transparency requirements in financial AI systems.

3. Reduction of False Positive Alerts. The coupled evaluation of emotional hype signals, order book imbalance, abnormal trading volume, order flow imbalance, and liquidity shock indicators within the extended five-signal manipulation risk model reduces the false alert rate characteristic of single-modality anomaly detectors. The integration of multiple complementary signals improves the robustness and specificity of manipulation detection.

4. Adaptive Operation Across Market Regimes. The Regime-Adaptive Multimodal Attention mechanism dynamically adjusts the weighting of all input signals according to the currently detected market regime. This enables the system to operate effectively across normal, volatile, event-driven, and manipulation-risk market conditions without requiring manual parameter reconfiguration.

5. Protection of Retail Investors. The invention generates interpretable alerts and monitoring outputs communicated to retail investors through an accessible asynchronous monitoring interface. By informing users about potential manipulation risks in near real time and with counterfactual natural language explanations, the system contributes directly to improved investor protection and market integrity.

6. Enhanced Decision Support for Financial Monitoring Systems. The integration of financial indicators, affective signals, market microstructure signals, and regime-aware multimodal reasoning produces a unified decision intelligence framework capable of supporting analysts, trading systems, algorithmic order management systems, and regulatory monitoring tools within a single consistent architecture.

6. DETAILED DESCRIPTION OF THE INVENTION

The following detailed description sets forth specific embodiments of the invention with reference to the accompanying drawings and mathematical formulations. This description is provided by way of illustration only and is not intended to limit the scope of the appended claims.

6.1 System Architecture Overview

The EMAFRS is organized as a modular pipeline architecture comprising six principal functional layers: the Data Acquisition Layer, the Preprocessing and Feature Engineering Layer, the Affective Signal Extraction Layer, the Market Regime Detection Engine, the Market Microstructure Intelligence Engine (MMIE), the Volatility-Aware Adaptive Fusion Engine, and the Explainable Decision Intelligence Output Layer. The MMIE constitutes a novel intermediary layer positioned between the regime classifier and the fusion engine, receiving the regime state R_t as a conditioning input and producing five microstructure feature signals that augment the manipulation risk model. Each layer is described in detail in the following sub-sections, and the interconnection of the layers is depicted in Figure 1.

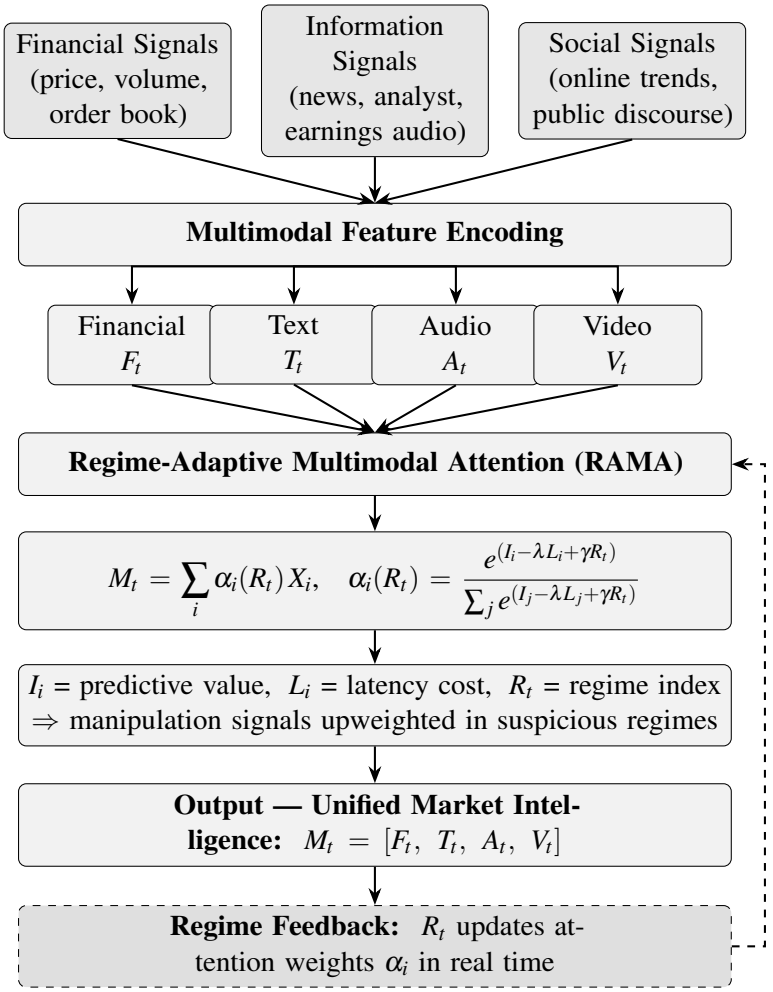


Figure 1: Diagram 1: Multimodal Market Observation Engine — regime-adaptive attention fusion producing unified market intelligence M_t.

6.2 Data Acquisition Layer

The Data Acquisition Layer (DAL) is responsible for the systematic ingestion of heterogeneous data from multiple sources, including structured numerical financial data feeds, derived algorithmic signal streams, and unstructured influencer communication channels.

6.2.1 Financial Market Time-Series Attributes

The DAL ingests primary market price data for each monitored financial instrument, characterized by the following attributes: date stamp, trading symbol, opening price, intraday high price, intraday low price, closing price, dividend-adjusted closing price, and aggregate trading volume. These attributes collectively constitute the

fundamental time-series representation of instrument-level market activity.

6.2.2 Return and Volatility Indicators

From the primary price data, the DAL computes and retains the following derived indicators: daily arithmetic return r_t, defined as $r_t = (P_t - P_{t-1})/P_{t-1}$; cumulative return over a rolling window; realized historical volatility; and maximum drawdown from a recent peak. These indicators provide a quantitative characterization of financial risk and return dynamics.

6.2.3 Technical Analysis Indicators

The system ingests a rich set of technical analysis indicators encompassing trend, momentum, and mean-reversion signals. Specifically, the following indicators are incorporated: Simple Moving Averages over 10-day and 50-day windows (SMA_{10}, SMA_{50}); Exponential Moving Averages over 12-day and 26-day windows (EMA_{12}, EMA_{26}); the Relative Strength Index (RSI) measuring momentum on a normalized scale; the Moving Average Convergence Divergence indicator (MACD), its signal line, and the MACD histogram; and the upper and lower Bollinger Band envelopes together with the normalized band position indicator. These indicators represent the primary computational substrates of classical technical analysis and serve as core quantitative features for the fusion engine.

6.2.4 Algorithmic Intelligence Signals and Adaptive Weights

The DAL further ingests the outputs of ancillary algorithmic scoring systems. These include the Adaptive Market Signal Fusion (AMSF) score and binary signal, the Fundamental Technical Composite Score (FTCS) and its generating formula, a composite Signal Confidence score, and Signal Confidence Entropy representing the distributional uncertainty of the confidence estimate. These algorithmic outputs are treated as meta-signals providing a second-order characterization of the quantitative evidence base. Additionally, the system processes Adaptive Signal Weighting parameters (ω_{MACD}, ω_{MA}, ω_{RSI}, ω_{BB}) that govern the relative contribution of each technical indicator category within the algorithmic scoring framework.

6.2.5 Portfolio and Risk Management Indicators

The risk dimension of the data acquisition pipeline includes: Dynamic Position Size (DPS) parameters capturing allocation strategies; Conditional Value at Risk (CVaR) representing tail risk exposure at a specified confidence level; Residual Volatility capturing idiosyncratic risk unexplained by systematic factors; and rolling Value-at-Risk (VaR) change metrics. These indicators enable the fusion engine to contextualize predictions within a risk-adjusted portfolio management framework.

6.2.6 High-Frequency Microstructure Signals

A critical and technically distinctive data acquisition capability of the present invention is the ingestion of microsecond-resolution market microstructure signals that are absent from all prior art multimodal financial AI systems. These signals include the following.

Order Book Depth and Imbalance. The system acquires Level 2 and Level 3 order book snapshots at tick resolution, from which it computes the Order Book Imbalance metric:

$$\mathrm{OBI}_t = \frac{V_t^{\mathrm{bid}} - V_t^{\mathrm{ask}}}{V_t^{\mathrm{bid}} + V_t^{\mathrm{ask}}}$$

where V_t^{bid} and V_t^{ask} denote the cumulative bid-side and ask-side volume within a specified price range. Extreme values of OBI_t signal structural liquidity imbalances characteristic of the pre-execution phase of pump-and-dump schemes.

Liquidity Withdrawal Indicators. The system monitors the rate of change of quoted depth across price levels to detect sudden liquidity withdrawal events, which are a microstructural precursor to manipulated price movements. A liquidity withdrawal signal is constructed as:

$$\mathrm{LW}_t = -\frac{d}{dt}\sum_{\ell} D_t^{(\ell)}$$

where $D_t^{(\ell)}$ is the quoted depth at price level ℓ.

Trade Flow and Toxicity Indicators. The system computes the Volume-Synchronized Probability of Informed Trading (VPIN) as a measure of order flow toxicity, capturing the fraction of volume attributable to informed directional trading as distinct from uninformed liquidity provision.

Alpha Decay and Persistence Monitoring. The system monitors alpha signal decay

dynamics using an exponential persistence model:

$$\alpha_t^{\text{decay}} = \alpha_t \cdot e^{-\lambda_\alpha \Delta t}$$

where λ_α is the alpha decay rate estimated from rolling out-of-sample performance and Δt is the elapsed time since signal generation. Rapid alpha decay under elevated emotional hype conditions is treated as a manipulation risk indicator, as artificially inflated signals exhibit faster-than-normal decay when the manipulative hype subsides.

Execution Latency Tracking. The system records the computational latency L_i associated with each input signal channel, enabling the Regime-Adaptive Multimodal Attention mechanism to penalize slow signals during high-frequency decision windows and to support latency-constrained microsecond-scale monitoring.

6.2.7 Market Entropy and Information Dynamics

The system additionally acquires Market Entropy measurements that quantify the information complexity and disorder present in the prevailing market microstructure; the Hurst Exponent characterizing the long-memory and trend-persistence properties of return series; the Regime Adaptive Behavioral Estimator (RABE) regime classification signal; Decay Factor and Decayed Alpha parameters capturing temporal information decay in adaptive signal weighting; Wasserstein Orthogonality metrics quantifying distributional separation between market states; and Liquidity-Adjusted Cost metrics accounting for transaction cost impacts on strategy performance.

6.3 Preprocessing and Feature Engineering Layer

The Preprocessing and Feature Engineering Layer applies standardized data quality, normalization, and transformation operations to the raw inputs acquired by the DAL prior to downstream processing.

All numerical financial and indicator variables undergo z-score standardization within rolling temporal windows to ensure stationarity and comparability across instruments and time periods. Time-series data are aligned to a common temporal index using forward-fill interpolation to address gaps arising from market closure periods.

Textual influencer communication data undergo a multi-stage preprocessing pipeline

comprising Unicode normalization, punctuation and special-character sanitization, tokenization using subword tokenization consistent with the pre-trained transformer models employed downstream, stop-word filtering appropriate for financial discourse, and conversion to dense continuous-vector semantic embeddings.

A volatility estimation sub-module computes current realized and implied volatility estimates and assigns each observation to a discrete volatility regime label (Low, Medium, High, or Extreme) using a Hidden Markov Model trained on historical volatility time series, as described in Section 6.5.

6.4 Affective Signal Extraction Module

The Affective Signal Extraction Module (ASEM) constitutes a core inventive component of the present disclosure. The ASEM is designed to analyze the communications of identifiable market influencers and to extract structured affective feature vectors representing the emotional and sentiment-laden informational content of those communications.

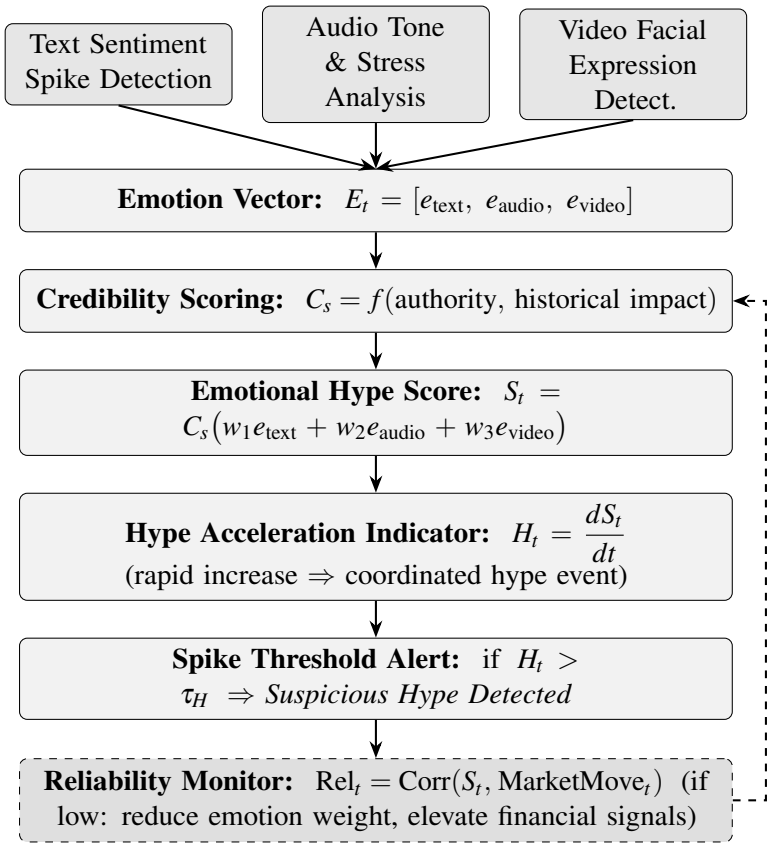


Figure 2: Diagram 2: Market Sentiment and Hype Detection Engine — emotion extraction, credibility scoring, and hype acceleration detection.

6.4.1 Influencer Stratification

The ASEM recognises that market influencers occupy differentiated institutional roles that modulate the market impact and interpretive context of their communications. The system therefore stratifies influencers into the following categories, each associated with role-specific processing parameters:

Government Policymakers communicate through legislative sessions, budget announcements, press conferences, and official statements relating to fiscal, regulatory, and trade policy. Their communications carry macroeconomic policy signal content with high market-wide impact.

Central Bank and Monetary Authority Officials communicate through monetary policy meeting minutes, forward guidance statements, speeches, and testimony relating to interest rate policy, quantitative easing, and financial stability assessment. Central bank communications are processed with particular attention to policy uncertainty signaling.

Corporate Executives and Institutional Representatives communicate through earnings calls, investor day presentations, analyst briefings, and media interviews. Their communications carry firm-specific and sector-level signal content.

Institutional Investors and Analysts communicate through research reports, media commentary, and public disclosures. Their communications carry information about professional consensus and analytical opinion.

Social Media Participants and Retail Investors communicate through platforms including financial discussion forums, micro-blogging services, and broadcast comment streams. Their communications capture retail sentiment dynamics and information diffusion patterns.

6.4.2 Emotion Detection Sub-Network

The Emotion Detection Sub-network (EDSN) applies a multi-label emotion classification model to pre-processed influencer communications to assign quantitative scores across a predefined set of discrete emotion categories. In the preferred embodiment, the emotion category set comprises: *Confidence*, *Uncertainty*, *Optimism*, *Fear*, *Caution*, *Assertiveness*, and *Evasiveness*. These categories are selected specifically for their financial market relevance, as they correspond to the principal affective states observed in market-relevant influencer discourse.

The EDSN is implemented as a fine-tuned transformer-based language model with a multi-label classification head appended. Let $\mathbf{e}_t = (e_1^t, e_2^t, \ldots, e_K^t) \in [0,1]^K$ denote the output emotion score vector for influencer communication at time t, where K is the number of emotion categories and each e_k^t represents the posterior probability of emotion category k being present in the communication.

6.4.3 Sentiment Polarity Classifier

The Sentiment Polarity Classifier (SPC) produces a signed scalar sentiment score $s_t \in [-1,+1]$ representing the overall directional sentiment orientation (bearish to bullish) of an influencer communication. The SPC is implemented as a fine-tuned transformer model with a regression head, trained on a domain-specific corpus of labelled financial communications. For social media inputs, the SPC additionally produces an intensity weight $\iota_t \in [0,1]$ proportional to the estimated reach and engagement metrics of the communication channel.

6.4.4 Contextual Semantic Encoder

The Contextual Semantic Encoder (CSE) produces a dense d_c-dimensional semantic embedding vector $\mathbf{c}_t \in \mathbb{R}^{d_c}$ capturing the topical and contextual content of the influencer communication. The CSE output represents the semantic relationship of the communication to financial concepts including monetary policy, trade conditions, earnings expectations, regulatory change, and systemic risk, providing a rich contextual substrate for downstream fusion.

6.4.5 Affective Feature Vector Construction

The final affective feature vector for influencer communication at time t is constructed as:

$$\mathbf{a}_t = \left[\mathbf{e}_t \parallel s_t \parallel \iota_t \parallel \mathbf{c}_t \parallel \rho_t\right] \tag{1}$$

where $\parallel$ denotes vector concatenation and $\rho_t \in \{1,\ldots,N_r\}$ is the one-hot encoded influencer role category index with $N_r = 5$ role categories. The resulting affective feature vector $\mathbf{a}_t \in \mathbb{R}^{d_a}$ thus encodes emotional scores, sentiment polarity, engagement intensity, semantic context, and influencer role identity in a unified structured representation.

6.5 Market Regime Detection Engine

The Market Regime Detection Engine (MRDE) identifies the prevailing structural state of financial market dynamics and outputs a discrete regime label $R_t \in \{1,2,3,4\}$ corresponding to the four volatility regimes: Low ($R = 1$), Medium ($R = 2$), High ($R = 3$), and Extreme ($R = 4$).

Regime classification is performed by a Hidden Markov Model (HMM) with Gaussian emission distributions, parameterized by the following observable state variables: the 21-day realized volatility estimate $\hat{\sigma}_t$, the Market Entropy measure H_t, the Hurst Exponent $\mathscr{H}_t$, and the RABE regime signal. The HMM transition probability matrix $\mathbf{A}$ and emission parameters are estimated by the Baum-Welch algorithm applied to a historical training window.

The Hurst Exponent $\mathscr{H}_t$ is estimated using the rescaled range statistic:

$$\mathscr{H}_t = \frac{\log\big(R(n)/S(n)\big)}{\log(n)} \tag{2}$$

where $R(n)$ denotes the range of the cumulative deviation series over window n and $S(n)$ denotes the standard deviation of the return series. Values $\mathscr{H} > 0.5$ indicate persistent trending behavior; values $\mathscr{H} < 0.5$ indicate mean-reverting dynamics; and $\mathscr{H} \approx 0.5$ indicates a random walk consistent with market efficiency.

Market Entropy is computed as the Shannon entropy of the empirical return distribution over a rolling estimation window:

$$H_t = -\sum_{b=1}^{B} p_b \log_2 p_b \tag{3}$$

where p_b represents the probability mass assigned to return bin b in a histogram with B bins constructed from the returns within the estimation window. Elevated entropy indicates distributional uncertainty and is associated with elevated volatility regimes.

6.6 Market Microstructure Intelligence Engine

The Market Microstructure Intelligence Engine (MMIE) constitutes a dedicated processing layer positioned between the Market Regime Detection Engine and the Volatility-Aware Adaptive Fusion Engine in the EMAFRS processing pipeline. The MMIE operates on tick-resolution and millisecond-scale order book data to

extract structural market microstructure features that are orthogonal to the high-level regime classification produced by the MRDE and the affective signals produced by the ASEM. The outputs of the MMIE are incorporated directly into the extended manipulation risk model, enabling the system to detect both the behavioral and structural signatures of coordinated market manipulation simultaneously.

The position of the MMIE in the processing pipeline is:

Market Regime Detection $\longrightarrow$ Microstructure Intelligence (MMIE) $\longrightarrow$ Manipulation Risk Model

6.6.1 Order Book Imbalance

The Order Book Imbalance (OBI) metric quantifies the directional pressure imbalance between the bid and ask sides of the limit order book at any point in time. It is computed as:

$$\mathrm{OBI}_t = \frac{V_t^{\mathrm{bid}} - V_t^{\mathrm{ask}}}{V_t^{\mathrm{bid}} + V_t^{\mathrm{ask}}} \tag{4}$$

where V_t^{bid} and V_t^{ask} denote the cumulative quoted volume on the bid and ask sides respectively within a configurable price depth window. The value OBI_t lies in $[-1,+1]$; values near $+1$ indicate overwhelming buy-side pressure consistent with coordinated buying campaigns characteristic of the accumulation phase of pump-and-dump schemes; values near -1 indicate aggressive sell-side pressure consistent with the distribution or dump phase.

In the context of manipulation detection, the system monitors not only the instantaneous OBI_t value but also its rate of change:

$$\Delta\mathrm{OBI}_t = \mathrm{OBI}_t - \mathrm{OBI}_{t-\delta} \tag{5}$$

Sudden large positive jumps in $\Delta\mathrm{OBI}_t$ coinciding with elevated emotional hype scores S_t constitute a primary composite indicator of coordinated pump activity.

6.6.2 Liquidity Shock Indicator

The Liquidity Shock Indicator (LSI) detects sudden and abnormal widening of the bid-ask spread relative to its recent historical baseline, which is a microstructural

precursor to price dislocation during manipulation events:

$$\mathrm{LS}_t = \frac{\left|\mathrm{Spread}_t - \overline{\mathrm{Spread}}\right|}{\overline{\mathrm{Spread}}} \tag{6}$$

where $\mathrm{Spread}_t = P_t^{\mathrm{ask}} - P_t^{\mathrm{bid}}$ is the current quoted spread and $\overline{\mathrm{Spread}}$ is the rolling mean spread computed over a baseline estimation window. Elevated LS_t values indicate that market makers are withdrawing liquidity in response to perceived information asymmetry or directional risk, a pattern consistently observed in the late accumulation and early distribution phases of pump-and-dump events.

The system additionally monitors the *depth depletion rate* at each price level:

$$\mathrm{DDR}_t^{(\ell)} = -\frac{d}{dt} D_t^{(\ell)} \tag{7}$$

where $D_t^{(\ell)}$ is the quoted depth at price level ℓ. Synchronized depth depletion across multiple price levels simultaneously is a characteristic microstructural signature of quote stuffing and spoofing strategies that frequently accompany coordinated manipulation.

6.6.3 Order Flow Imbalance and Toxicity

The Order Flow Imbalance (OFI) metric measures the net directional aggression of executed trades over a rolling observation window:

$$\mathrm{OFI}_t = V_t^{\mathrm{buy,executed}} - V_t^{\mathrm{sell,executed}} \tag{8}$$

where $V_t^{\mathrm{buy,executed}}$ and $V_t^{\mathrm{sell,executed}}$ denote the total volume of marketable buy and sell orders executed within the window. Large positive spikes in OFI_t relative to baseline indicate a coordinated and aggressive buying campaign inconsistent with organic market activity. Combined with the emotional hype acceleration indicator $H_t = dS_t/dt$, elevated OFI_t provides a powerful confirmation signal for pump-phase detection.

The system further estimates order flow toxicity using the Volume-Synchronized Probability of Informed Trading (VPIN):

$$\mathrm{VPIN}_t = \frac{\sum_{\tau=1}^{n} \left|V_\tau^{\mathrm{buy}} - V_\tau^{\mathrm{sell}}\right|}{\sum_{\tau=1}^{n} V_\tau} \tag{9}$$

where the summation is taken over n equal-volume buckets. Elevated VPIN values indicate that a disproportionate fraction of traded volume is being directed by informed or manipulative actors, consistent with the coordinated execution patterns of pump-and-dump operations.

6.6.4 Alpha Persistence Monitor

The Alpha Persistence Monitor (APM) evaluates whether the trading signals generated by the system exhibit the temporal stability characteristic of fundamentally-driven alpha, or the rapid deterioration characteristic of hype-inflated artificial signals. Alpha persistence is quantified as the rolling cross-correlation between the generated signal and subsequent realized returns:

$$\mathrm{AP}_t = \mathrm{Corr}\big(\mathrm{signal}_t,\ r_{t+\Delta}\big) \tag{10}$$

where $r_{t+\Delta}$ denotes the realized return at lag Δ following signal generation. A rapid collapse in AP_t — defined as a decrease exceeding a threshold δ_{AP} within a rolling window — is interpreted as evidence that the signal was driven by hype rather than fundamental information, since artificially inflated prices revert rapidly once the manipulation subsides.

The system monitors the alpha decay dynamics using:

$$\alpha_t^{\mathrm{decay}} = \alpha_t \cdot e^{-\lambda_\alpha \Delta t} \tag{11}$$

where λ_α is the alpha decay rate estimated adaptively from rolling out-of-sample performance measurement. Divergence between the raw signal strength and the decay-adjusted signal provides a continuous manipulation-likelihood indicator.

6.6.5 Extended Manipulation Risk Model

The MMIE outputs are incorporated into an extended manipulation risk scoring function that jointly evaluates behavioral, structural, and microstructural manipulation evidence:

$$\mathrm{Risk}_t = w_1 S_t + w_2\,\mathrm{OBI}_t + w_3\,\mathrm{Vol}_{\mathrm{anomaly}} + w_4\,\mathrm{OFI}_t + w_5\,\mathrm{LS}_t \tag{12}$$

where S_t is the emotion-weighted hype score capturing behavioral manipulation signals; OBI_t captures order book directional pressure; $\text{Vol}_{\text{anomaly}}$ captures abnormal volume deviations from baseline; OFI_t captures order flow aggression; and LS_t captures liquidity withdrawal dynamics. The weights $w_1, \ldots, w_5$ are learned during training and are further modulated by the current regime classification R_t, such that microstructure weights are elevated during manipulation-risk regimes.

When $\text{Risk}_t > \tau$, the system flags possible pump-and-dump activity and activates the Counterfactual Citizen Alert generation pipeline. The extended five-signal formulation is technically superior to prior art single-signal or dual-signal manipulation detectors because it captures manipulation at multiple structural levels simultaneously: the behavioral and communicative level through S_t; the order book structural level through OBI_t and LS_t; and the execution flow level through OFI_t and $\text{Vol}_{\text{anomaly}}$. No prior art system integrates all five signal categories within a single unified risk model.

6.7 Volatility-Aware Adaptive Fusion Engine

The Volatility-Aware Adaptive Fusion Engine (VAFE) constitutes the central computational component of the EMAFRS, integrating quantitative financial feature vectors with extracted affective feature vectors under the modulating influence of the current market regime and volatility state.

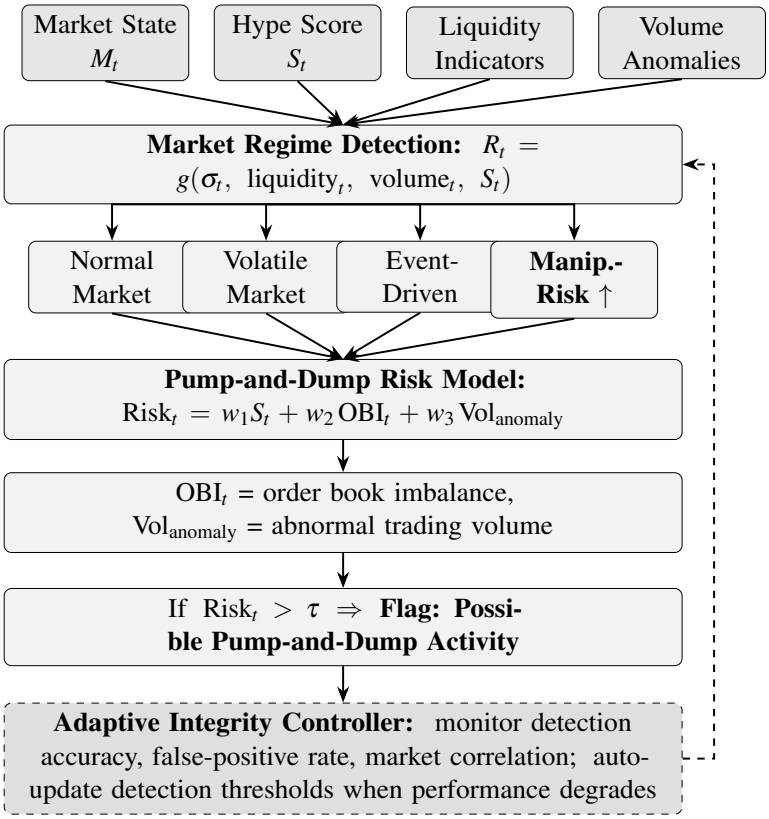


Figure 3: Diagram 3: Manipulation-Aware Market Regime Detection — regime classification and pump-and-dump risk scoring with adaptive feedback.

6.7.1 Quantitative Feature Vector

The quantitative feature vector at time t is defined as:

$$\mathbf{q}_t = \left[\tilde{\mathbf{x}}_t^{\text{price}} \parallel \tilde{\mathbf{x}}_t^{\text{tech}} \parallel \tilde{\mathbf{x}}_t^{\text{risk}} \parallel \tilde{\mathbf{x}}_t^{\text{algo}}\right] \in \mathbb{R}^{d_q} \tag{13}$$

where $\tilde{\mathbf{x}}^{\text{price}}$, $\tilde{\mathbf{x}}^{\text{tech}}$, $\tilde{\mathbf{x}}^{\text{risk}}$, and $\tilde{\mathbf{x}}^{\text{algo}}$ denote the standardized price, technical indicator, risk management, and algorithmic signal sub-vectors respectively.

6.7.2 Adaptive Signal Weighting Mechanism

The primary inventive contribution of the VAFE resides in its adaptive signal weighting mechanism, which determines the relative contribution of quantitative versus affective signals to the fused representation as a function of the current volatility regime and market entropy state.

Let $\lambda_t \in [0,1]$ denote the adaptive weight assigned to the quantitative feature compo-

nent at time t. The adaptive weight is computed as:

$$\lambda_t = \sigma\big(\mathbf{w}_\lambda^\top \phi_t + b_\lambda\big) \tag{14}$$

where $\sigma(\cdot)$ is the sigmoid activation function, $\mathbf{w}_\lambda$ and b_λ are learnable parameters, and ϕ_t is the regime conditioning vector:

$$\phi_t = \big[\, R_t \parallel H_t \parallel \hat{\sigma}_t \parallel \mathscr{H}_t \parallel \Delta H_t \,\big] \tag{15}$$

with $\Delta H_t = H_t - H_{t-1}$ representing the rate of change in market entropy. When $\lambda_t \to 1$, the fusion engine prioritizes quantitative financial signals, consistent with behavior in low-volatility, trend-persistent regimes. When $\lambda_t \to 0$, the engine elevates affective signals, consistent with behavior in high-volatility, event-driven regimes where behavioral and communicative signals dominate price discovery.

The fused representation is constructed as:

$$\mathbf{z}_t = \big[\, \lambda_t \cdot \mathbf{q}_t \parallel (1-\lambda_t) \cdot \mathbf{a}_t \,\big] \in \mathbb{R}^{d_q + d_a} \tag{16}$$

6.7.3 Volatility-Aware Inference Formulation

The multimodal reasoning network f_θ maps the fused representation $\mathbf{z}_t$ to a joint prediction of market direction and risk estimate. The volatility-aware inference objective is formulated as the joint minimization of a prediction loss and a risk calibration penalty:

$$\mathscr{L}_{\text{VAFE}} = \mathscr{L}_{\text{pred}}\big(f_\theta(\mathbf{z}_t), y_t\big) + \mu_R \cdot \mathscr{L}_{\text{risk}}\big(g_\phi(\mathbf{z}_t), \text{CVaR}_t\big) + \eta \cdot \Omega(\theta, \phi) \tag{17}$$

where $y_t \in \{-1, 0, +1\}$ is the realized market direction label, CVaR_t is the empirical Conditional Value at Risk computed from realized returns, μ_R is a regime-dependent risk penalty weight satisfying $\mu_R \propto R_t$, Ω is a regularization term, and η is the regularization coefficient. The regime-dependent weighting of the risk penalty ensures that the model is increasingly penalized for risk miscalibration in high-volatility regimes.

The full volatility-aware market inference formulation is expressed as:

$$P(y_t \mid \mathbf{z}_t, R_t) = \text{softmax}\Big(f_\theta(\mathbf{z}_t) \cdot \gamma(R_t)\Big) \tag{18}$$

where $\gamma(R_t) \in \mathbb{R}^+$ is a temperature scaling parameter that is a monotonically decreasing function of the regime index R_t, producing calibrated posterior distributions with increased uncertainty in high-volatility regimes consistent with epistemic humility under market stress conditions.

6.8 Explainability Module

The Explainability Module (XAI Layer) generates human-interpretable attribution explanations for each prediction produced by the VAFE. The module employs a combination of gradient-based attribution methods, attention weight visualization, and natural language generation to produce multi-level explanations accessible to practitioners of varying technical sophistication.

6.8.1 Feature Attribution via Integrated Gradients

For each prediction, the contribution of each input feature $z_{t,j}$ to the output $\hat{y}_t$ is quantified using the Integrated Gradients attribution method:

$$\mathrm{IG}_j(z_t) = (z_{t,j} - z^0_{t,j}) \cdot \int_{\alpha=0}^{1} \frac{\partial f_\theta(\mathbf{z}^0_t + \alpha(\mathbf{z}_t - \mathbf{z}^0_t))}{\partial z_{t,j}} \, \mathrm{d}\alpha \tag{19}$$

where $\mathbf{z}^0_t$ is a baseline input vector set to the time-series mean of the fused feature representation. The attribution scores $\{\mathrm{IG}_j\}$ satisfy the completeness property, ensuring that attributions sum to the difference between the prediction at the actual input and the prediction at the baseline, providing a theoretically grounded decomposition of model output.

6.8.2 Attention-Based Explanation for Affective Signals

For the affective signal component, the transformer attention mechanism provides a secondary explanation modality. The module identifies the specific influencer communication tokens receiving the highest attention weight in the prediction layer, associating specific linguistic sub-expressions with predictive influence. This capability enables the system to generate fine-grained explanations of the form: the predicted downward market movement is partially attributable to the *uncertainty* and *caution* emotion signals extracted from a specific central bank statement.

6.8.3 Natural Language Explanation Generation

The explainability module includes a template-driven natural language generation component that translates the quantitative attribution scores and attention highlights into structured textual explanations composed of the following standardized elements: a market direction summary statement; a ranked list of the top contributing financial indicators with their attribution magnitudes; a ranked list of the top contributing affective signals with their associated influencer categories; the current market regime and its influence on signal weighting; and a confidence assessment qualified by the current volatility state.

6.9 Decision Intelligence Output Layer

The Decision Intelligence Output Layer aggregates the outputs of the VAFE and the Explainability Module into a structured decision packet comprising the following elements:

The *Predicted Market Trend Direction* output provides a ternary classification (bullish, neutral, or bearish) accompanied by posterior probability estimates for each class.

The *Volatility Alert* output provides a volatility regime classification and directional volatility forecast, enabling practitioners to adjust position sizing and risk exposure pre-emptively.

The *Sentiment-Driven Signal* output provides a signed sentiment contribution score representing the net directional influence of all ingested affective signals on the prediction.

The *Confidence Score* output provides a calibrated probability estimate associated with the predicted market direction, adjusted for current volatility regime uncertainty.

The *Explanation Summary* output provides the structured natural language attribution summary generated by the Explainability Module, fulfilling the interpretability requirement of the invention.

The *Risk-Adjusted Decision Intelligence* output integrates the market direction prediction with the CVaR risk estimate to produce a risk-adjusted signal suitable for use in portfolio allocation models.

The *Alpha Stability Signal* output reports the current alpha persistence score α_t^{decay}

and flags conditions under which hype-driven signal inflation may be causing rapid alpha decay, enabling execution systems to reduce position size or pause strategy deployment when detected alpha is likely to be manipulative rather than fundamental.

The *Execution-Aware Alert* output translates the manipulation risk score and microstructure signals into specific order execution recommendations, including guidance on order size reduction during detected liquidity withdrawal events, routing to alternative execution venues during suspected manipulation regimes, and queue position adjustment in response to detected order book imbalance. This execution-aware output layer directly interfaces with algorithmic order management systems, enabling the manipulation detection capability of the present invention to influence live trading execution decisions in a latency-constrained environment.

The *Microstructure Integrity Report* output provides a structured summary of order book health, liquidity conditions, and trade flow toxicity at the time of each manipulation alert, enabling post-trade analysis and regulatory reporting of suspected manipulation events with full microstructure evidence.

7. BRIEF DESCRIPTION OF THE DRAWINGS

The invention is illustrated with reference to the accompanying drawings submitted as separate drawing sheets. The drawings are black and white line diagrams with all components numbered. Like reference numerals denote corresponding parts throughout the figures.

FIG. 1 — Multimodal Market Observation Engine (System Architecture)

FIG. 1 depicts the overall system architecture of the AI Market Integrity Framework (AMIF). Reference numerals denote: (101) Financial Signal Input; (102) Text/News Signal Input; (103) Audio Signal Input; (104) Video Signal Input; (105) Multimodal Feature Encoding Layer; (106) Regime-Adaptive Multimodal Attention (RAMA) Fusion Module; (107) Market State Vector M_t; (108) Regime Feedback Controller.

FIG. 2 — Multimodal Affective Market Influence Engine

FIG. 2 depicts the emotion extraction and hype detection pipeline. Reference numerals denote: (201) Influencer Communication Inputs; (202) Text Sentiment Spike Detector; (203) Audio Tone and Stress Analyser; (204) Video Facial Expression Recogniser; (205) Emotion Vector E_t; (206) Speaker Credibility Scoring Module;

(207) Emotional Hype Score S_t; (208) Hype Acceleration Indicator H_t; (209) Reliability Monitor Feedback Loop.

FIG. 3 — Manipulation-Aware Market Regime Detection and Microstructure Intelligence Engine

FIG. 3 depicts the regime detection and Market Microstructure Intelligence Engine (MMIE). Reference numerals denote: (301) Signal Aggregation Layer; (302) Hidden Markov Model Regime Classifier; (303) Regime Output R_t; (304) Order Book Imbalance Module OBI_t; (305) Liquidity Shock Indicator LS_t; (306) Order Flow Imbalance Module OFI_t; (307) Alpha Persistence Monitor AP_t; (308) Extended Manipulation Risk Scorer Risk_t; (309) Threshold Comparator τ; (310) Adaptive Integrity Controller Feedback.

FIG. 4 — Explainable Citizen Alert and Human-in-the-Loop Control Engine

FIG. 4 depicts the explainability and citizen alert subsystem. Reference numerals denote: (401) Prediction Output α_t; (402) Feature Attribution Engine; (403) Counterfactual Comparison Module; (404) Natural Language Alert Generator; (405) Citizen Monitoring Interface; (406) Risk Alert Delivery; (407) Suspicious Asset Tracker; (408) Explanation Review Panel; (409) Global Adaptive Integrity Controller.

8. DRAWINGS

The following figures accompany this specification and form an integral part thereof. Figures 1 through 4 have been incorporated inline within the Detailed Description of the Invention as Figures 1 through 3. Additional drawing representations are provided below for illustrative clarity.

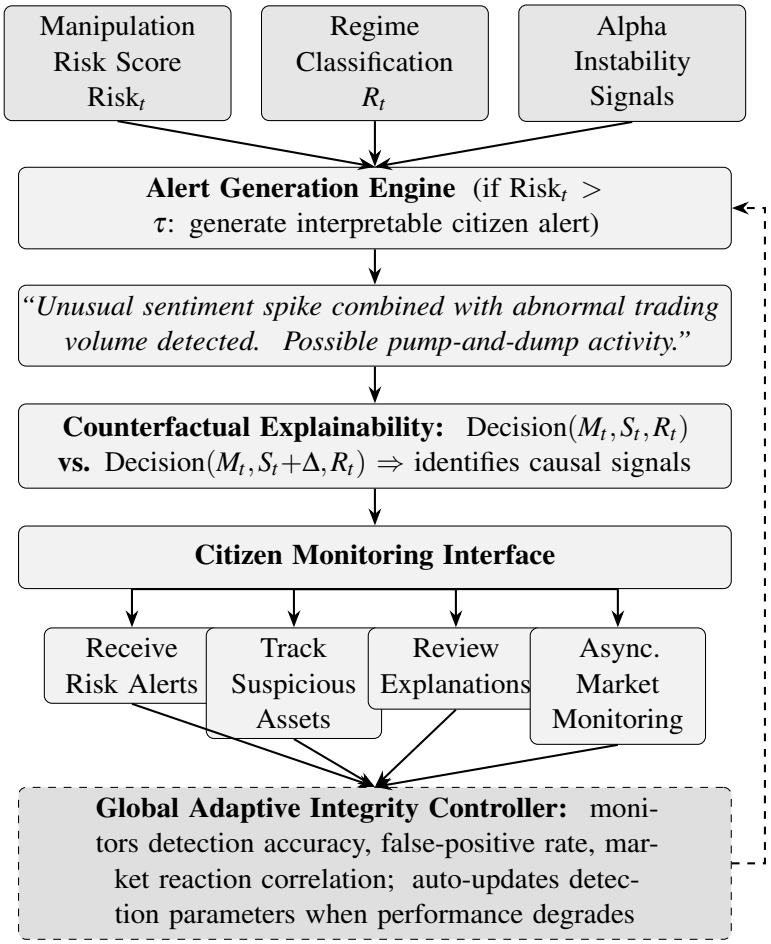


Figure 4: Diagram 4: Explainable Citizen Alert and Monitoring System — counterfactual explainability, interpretable alerts, and asynchronous citizen interface.

9. STATEMENT OF INVENTIVE STEP AND NON-OBVIOUSNESS

The present invention satisfies the non-obviousness requirement on the following grounds.

9.1 Novelty of Individual Components

The Regime-Adaptive Multimodal Attention mechanism introduces a novel parameterisation in which the regime index R_t appears as an explicit additive term in the softmax exponent governing cross-modal attention weights. No prior art multimodal attention mechanism conditions fusion weights on a concurrently computed market regime classification in this manner.

The Market Microstructure Intelligence Engine (MMIE) introduces a novel dedicated

processing layer that operates on tick-resolution order book data to compute four structurally distinct microstructure signals — Order Book Imbalance (OBI_t), Liquidity Shock Indicator (LS_t), Order Flow Imbalance (OFI_t), and Alpha Persistence Monitor (AP_t) — and positions this layer explicitly between the regime classifier and the fusion engine so that microstructure evidence is available to the manipulation risk model before the fusion representation is assembled. No prior art multimodal financial AI system includes a dedicated microstructure intelligence layer in this architectural position.

The Emotion-Regime-Microstructure Coupled Pump-and-Dump Detection model extends the prior art dual-signal risk model to a five-signal formulation: $\mathrm{Risk}_t = w_1 S_t + w_2 \mathrm{OBI}_t + w_3 \mathrm{Vol}_{\mathrm{anomaly}} + w_4 \mathrm{OFI}_t + w_5 \mathrm{LS}_t$. This coupling of cross-modal affective signals with four independent microstructure signals within a single unified risk scoring function is novel. No prior art manipulation detection system integrates all five signal categories simultaneously.

The Counterfactual Citizen Alert Explainability mechanism generates natural language manipulation alerts derived from counterfactual reasoning over multimodal input signals. No prior art financial alert system generates counterfactual explanations in natural language targeted at non-expert retail investor recipients in the context of manipulation detection.

9.2 Non-Obviousness of the Combined Architecture

The combined closed-loop architecture of the present invention is non-obvious for the following reasons. A person skilled in the art of financial machine learning, confronted with the technical problem of detecting emotion-driven market manipulation and alerting retail investors, would not have arrived at the specific combination of: regime-conditioned multimodal attention; emotion-regime coupled risk scoring; and counterfactual natural language alert generation. The reason is that the prior art treats these as problems in separate technical domains — multimodal fusion, anomaly detection, and explainability respectively — and no prior art teaching suggests that the output of a regime classifier should dynamically modulate multimodal attention weights, or that cross-modal affective signals should be coupled with structural market microstructure metrics in a shared risk function, or that counterfactual reasoning over the resulting multimodal inputs should be used to generate citizen alerts.

9.3 Non-Obviousness Arising from Microstructure Integration

A further and independent basis for non-obviousness resides in the integration of high-frequency market microstructure signals — including microsecond-resolution order book imbalance, liquidity withdrawal dynamics, order flow toxicity, and alpha decay monitoring — with cross-modal affective signals within the same unified risk scoring framework. High-frequency trading researchers and microstructure theorists have studied order book dynamics and alpha decay extensively as isolated technical problems. Multimodal AI researchers have studied sentiment fusion and emotion detection as separate technical problems. The insight that microstructure signals and affective signals jointly and synergistically characterize the pre-execution phase of coordinated market manipulation, and therefore must be coupled in a single risk model to achieve early detection, is not suggested by any prior art in either field and could not have been arrived at by ordinary combination of the prior art teachings of either community.

9.4 Technical Effect Produced

The interaction of the three inventive mechanisms, augmented by high-frequency microstructure signal integration, produces a measurable technical effect that is not achievable by their individual application: the ability to detect pump-and-dump manipulation patterns at the hype-generation stage, before price distortion is observable, by correlating emotional hype acceleration with concurrent order book imbalance and liquidity withdrawal microstructure signatures; to communicate the specific causal signals responsible for the detection to retail investors in natural language in real time; and to generate execution-aware alerts that can be consumed directly by algorithmic order management systems to protect both retail and institutional participants from manipulated market conditions. This technical effect directly addresses the protection of retail investors from coordinated market manipulation and constitutes a non-obvious technical contribution over the prior art.

Requirement	How Satisfied
Novelty	New RAMA formula with regime index in softmax exponent; new emotion-regime-microstructure coupled risk model; new counterfactual citizen alert mechanism; new alpha decay manipulation indicator
Non-Obviousness	Closed-loop interaction of three mechanisms; microstructure + affective signal coupling not suggested by any prior art in either HFT or multimodal AI domains; RAMA parameterisation is technically distinct from all static fusion architectures
Technical Effect	Early manipulation detection at hype-generation stage via OBI + affective coupling; microsecond-latency monitoring; execution-aware alerts for order management systems; real-time counterfactual citizen alerts
Industrial Applicability	High-frequency market surveillance platforms, retail investor protection systems, algorithmic trading risk management, securities regulatory tools, order execution management systems

Table 1: Summary of non-obviousness arguments.

9.5 Industrial Applicability and Section 3(k) Compliance

The present invention is a **deployable computer-implemented technical platform** with clear industrial applicability in securities market surveillance, retail investor protection, institutional trading risk management, and regulatory compliance monitoring. The system accepts physical market data feeds as input, applies multimodal processing across hardware and software modules, and produces actionable output alerts consumed by human users and automated trading systems. These technical effects — early manipulation detection, execution-aware microstructure alerts, counterfactual citizen notifications — are concrete and measurable outcomes of the deployed platform.

The mathematical formulations contained in this specification describe the computational operations performed by specific modules within the platform. They are **not claimed as mathematical methods, algorithms, or computer programs *per se*** within the meaning of Section 3(k) of the Patents Act, 1970. The claims are directed to the **technical system and its functional architecture**, not to the underlying mathematical relationships in the abstract. This distinction is analogous to the patentability of a technical apparatus that implements a physical process,

where the mathematical description of that process does not render the apparatus unpatentable. The present invention therefore falls outside the Section 3(k) exclusion and constitutes patentable subject matter under Indian patent law.

10. CLAIMS

1. A computer-implemented market surveillance platform and investor protection system configured to detect coordinated market manipulation and to deliver explainable citizen alerts, the platform comprising a plurality of interconnected hardware and software modules operating in combination to produce technical effects of early manipulation detection, reduced false alert rates, and real-time retail investor protection, the platform comprising: a multimodal data acquisition module configured to continuously ingest, from a plurality of heterogeneous real-time data sources, financial time-series signals, order book depth and dynamics, influencer text communications, audio speech streams, and video visual statements; a multimodal affective extraction module configured to analyze the ingested influencer communications across text, audio, and video modalities, to assign a credibility coefficient to each identified influencer source based on authority and historical market impact, and to produce a weighted emotional hype score representing the aggregate directional sentiment pressure attributable to influencer communications at each point in time; a manipulation-aware market regime detection engine configured to classify the prevailing market state into a plurality of discrete regime categories including a manipulation-risk regime, based on observed volatility, liquidity conditions, trading volume, and the emotional hype score; a Market Microstructure Intelligence Engine (MMIE) configured to compute, from tick-resolution order book data, a plurality of microstructure signals including an order book imbalance metric, a liquidity shock indicator, an order flow imbalance metric, and an alpha persistence score, and to produce an extended manipulation risk score integrating said microstructure signals with the emotional hype score and abnormal volume indicators; a Regime-Adaptive Multimodal Attention fusion module configured to dynamically adjust the relative weighting of each input signal modality as a function of the currently detected market regime, such that signals associated with market manipulation are automatically elevated in importance during suspicious market regimes; a counterfactual explainability module configured to generate human-interpretable natural language alerts by identifying the specific observed signals that caused the manipulation risk threshold to be exceeded and communicating those signals to retail investors

through an asynchronous monitoring interface; and a Global Adaptive Integrity Controller configured to monitor platform detection performance and to update detection parameters automatically when performance metrics degrade.

2. The system according to Claim 1, wherein the affective signal extraction module comprises a role-stratified processing pipeline that applies distinct natural language processing and emotion detection model configurations to each of a plurality of influencer role categories comprising government policymakers, central bank officials, corporate executives, institutional investors, and social media participants, such that the market impact and communicative context appropriate to each role category is reflected in the extracted affective feature vectors.

3. The system according to Claim 1, wherein the adaptive weighting coefficient λ_t is computed by a learned sigmoid function applied to a regime conditioning vector comprising the current volatility regime label, current market entropy, realized volatility estimate, Hurst Exponent, and rate of change of market entropy, such that the relative contribution of affective signals to the fused representation increases monotonically with increasing volatility regime and market entropy, thereby elevating behavioral and communicative signal integration during periods of heightened market stress and event-driven price discovery.

4. The system according to Claim 1, wherein the market regime detection engine classifies market states using hidden Markov model inference incorporating at minimum the following observable state variables: realized historical volatility over a rolling estimation window, Shannon entropy of the empirical return distribution, Hurst Exponent estimated by rescaled range analysis, and a regime adaptive behavioral estimator signal, wherein the resulting regime classification label modulates both the adaptive fusion weighting and a regime-dependent risk penalty weight in the training loss formulation.

5. The system according to Claim 1, wherein the explainability module employs Integrated Gradients attribution to compute per-feature contribution scores satisfying the completeness axiom, whereby the attribution scores sum to the difference between the model output at the actual fused input representation and the model output at a baseline representation, and wherein the attribution scores are further annotated with influencer role category identifiers for attribution scores derived from affective feature vector components, thereby enabling practitioner-accessible explanation of

the behavioral and quantitative drivers of each market prediction.

6. The system according to Claim 1, wherein the multimodal reasoning network generates calibrated posterior class probability distributions over predicted market direction by applying temperature scaling with a regime-dependent temperature parameter that decreases monotonically with increasing volatility regime index, such that the predictive confidence of the system is appropriately attenuated during high-volatility market regimes consistent with elevated epistemic uncertainty.

7. The system according to Claim 1, wherein the decision intelligence output layer produces a structured decision packet comprising at minimum: a ternary market direction classification with posterior probability estimates; a volatility regime alert and directional volatility forecast; a net sentiment contribution score derived from aggregated affective signals; a risk-adjusted decision signal integrating market direction prediction with Conditional Value at Risk estimate; and a natural language explanation summary generated by a template-driven language generation component mapping quantitative attribution scores and attention highlights to structured interpretable text identifying the dominant financial indicators, affective signals, influencer role categories, and market regime characteristics contributing to the prediction.

8. A computer-implemented method for generating explainable financial decision intelligence, the method comprising the steps of: acquiring, by a data acquisition module, heterogeneous input data comprising financial market time-series data, algorithmic indicator signals, and influencer communication data from a plurality of channels; extracting, by an affective signal extraction module, structured affective feature vectors from influencer communications using role-stratified natural language processing, emotion detection, sentiment polarity classification, and contextual semantic encoding; classifying, by a market regime detection engine, the current market state into a discrete volatility regime label using hidden Markov model inference over volatility, entropy, and persistence indicators; computing, by an adaptive fusion engine, a regime-conditioned adaptive weighting coefficient and constructing a fused feature representation that integrates weighted quantitative financial features and weighted affective features; generating, by a multimodal reasoning network, a market direction prediction and risk estimate from the fused feature representation under a volatility-aware joint optimization objective incorporating a regime-dependent risk penalty; and producing, by an explainability module, a structured attribution expla-

nation comprising Integrated Gradient feature attributions, attention-based affective signal highlights, and a natural language explanation summary, thereby enabling transparent, interpretable, and volatility-aware financial decision intelligence output.

9. The system according to Claim 1, wherein the Market Microstructure Intelligence Engine (MMIE) computes an Order Book Imbalance metric $\mathrm{OBI}_t = (V_t^{\mathrm{bid}} - V_t^{\mathrm{ask}})/(V_t^{\mathrm{bid}} + V_t^{\mathrm{ask}})$ and a rate-of-change indicator $\Delta\mathrm{OBI}_t = \mathrm{OBI}_t - \mathrm{OBI}_{t-\delta}$, wherein sudden positive jumps in $\Delta\mathrm{OBI}_t$ coinciding with elevated emotional hype scores S_t constitute a primary composite indicator of coordinated pump-phase activity.

10. The system according to Claim 1, wherein the MMIE computes a Liquidity Shock Indicator $\mathrm{LS}_t = |\mathrm{Spread}_t - \overline{\mathrm{Spread}}|/\overline{\mathrm{Spread}}$ quantifying abnormal bid-ask spread widening relative to a rolling baseline, and a depth depletion rate $\mathrm{DDR}_t^{(\ell)} = -dD_t^{(\ell)}/dt$ for each price level ℓ, wherein synchronized depth depletion across multiple price levels is classified as a structural precursor to manipulated price movements.

11. The system according to Claim 1, wherein the MMIE computes an Order Flow Imbalance metric $\mathrm{OFI}_t = V_t^{\mathrm{buy,executed}} - V_t^{\mathrm{sell,executed}}$ and a Volume-Synchronized Probability of Informed Trading (VPIN) score over equal-volume buckets, wherein elevated VPIN values in conjunction with elevated emotional hype scores indicate coordinated manipulative execution activity.

12. The system according to Claim 1, wherein the extended manipulation risk model evaluates a five-signal composite score $\mathrm{Risk}_t = w_1 S_t + w_2\,\mathrm{OBI}_t + w_3\,\mathrm{Vol}_{\mathrm{anomaly}} + w_4\,\mathrm{OFI}_t + w_5\,\mathrm{LS}_t$, wherein the weights w_1 through w_5 are jointly learned during training and are further modulated by the current regime classification R_t such that microstructure weights are automatically elevated during manipulation-risk regimes.

13. The system according to Claim 1, wherein the Alpha Persistence Monitor computes $\mathrm{AP}_t = \mathrm{Corr}(\mathrm{signal}_t, r_{t+\Delta})$ and monitors alpha decay dynamics via $\alpha_t^{\mathrm{decay}} = \alpha_t \cdot e^{-\lambda_\alpha \Delta t}$, wherein rapid collapse in AP_t beyond a threshold δ_{AP} within a rolling window is classified as evidence of hype-inflated rather than fundamental trading signal, activating a manipulation-likelihood flag.

14. The system according to Claim 1, wherein the counterfactual explainability module generates citizen alerts by computing the decision difference $\mathrm{Decision}(M_t, S_t, R_t) -$

Decision$(M_t, S_t + \Delta, R_t)$ across a plurality of hypothetical perturbation values Δ, and translates the resulting sensitivity profile into a ranked natural language explanation identifying the minimum signal change that would have prevented alert generation.

15. The system according to Claim 1, further comprising an execution-aware alert output layer configured to translate the manipulation risk score and microstructure signals into specific order execution recommendations including order size reduction during detected liquidity withdrawal events, routing to alternative execution venues during suspected manipulation regimes, and queue position adjustment in response to detected order book imbalance, wherein said output directly interfaces with algorithmic order management systems.

16. The system according to Claim 1, wherein the citizen monitoring interface delivers manipulation risk alerts to retail investors asynchronously through a push-notification mechanism, enabling investors who are not actively monitoring markets to receive interpretable warnings about suspicious market activity, including an explanation summary, the specific assets flagged, and the counterfactual condition that would have suppressed the alert.

17. The system according to Claim 1, further comprising a Global Adaptive Integrity Controller that continuously monitors system performance metrics including manipulation detection accuracy, false positive rate, and correlation between generated alerts and subsequent market movements, and automatically updates detection thresholds and model parameters when performance metrics fall below predefined bounds, ensuring robust operation as market manipulation strategies evolve over time.

18. The system according to Claim 1, wherein the Hype Acceleration Indicator $H_t = dS_t/dt$ is computed as a first-order temporal derivative of the emotion-weighted hype score, wherein values of H_t exceeding a threshold τ_H trigger a Spike Alert independently of the full five-signal manipulation risk score, providing an early-warning mechanism for rapidly developing coordinated hype events.

19. The system according to Claim 1, wherein the speaker credibility coefficient $C_s = f(\text{authority}, \text{credibility}, \text{historical impact})$ is computed for each influencer source using a learned function of the source authority tier, a time-decayed credibility score derived from historical correlation between the source's prior communica-

tions and subsequent market movements, and an event relevance score computed from semantic similarity between the communication content and current market conditions.

20. A computer-implemented financial intelligence system configured to detect market manipulation and generate interpretable alerts for users, the system comprising: a data ingestion engine configured to acquire heterogeneous financial market signals and communication signals from a plurality of sources, wherein the financial market signals include at least price, volume, and order book data, and wherein the communication signals include at least one of textual, audio, or video communications produced by market participants or influencers; a multimodal reasoning engine configured to combine the acquired quantitative market signals and the acquired behavioral communication signals to produce a manipulation risk estimate representing the likelihood of coordinated market manipulation activity at a given point in time; and an explainable alert generation engine configured to generate human-interpretable alerts that identify, in natural language, the specific signals that contributed to the manipulation risk estimate exceeding a threshold, and to deliver said alerts to users through an accessible monitoring interface; wherein the system produces a technical effect of enabling detection of market manipulation at an earlier stage than systems operating on financial signals alone, by integrating behavioral and communicative evidence with quantitative market evidence within a unified risk estimation framework.

21. The system of Claim 1, wherein the multimodal reasoning engine is further configured to classify a current market regime from a plurality of discrete regime states, and to dynamically adjust the relative weighting applied to each category of acquired signal as a function of the classified market regime, such that the contribution of behavioral communication signals to the manipulation risk estimate is automatically increased when the classified market regime indicates elevated manipulation risk, and automatically decreased when the classified market regime indicates normal market conditions.

22. The system of Claim 1, wherein the platform is deployed within at least one of: a securities exchange real-time monitoring system operated by an exchange operator or market regulator; an algorithmic trading risk management system operated by an institutional investor or trading firm; a financial regulatory surveillance system operated by a securities market authority; a retail investor protection application

delivering alerts to individual investors through a mobile or web-based interface; or a compliance monitoring platform operated by a financial institution subject to market integrity regulations; such that the manipulation detection and alert generation capabilities of the platform are operative regardless of the deployment context in which the platform is instantiated.

23. The system of Claim 1, wherein the data ingestion engine is further configured to acquire social media content and public broadcast content produced by identifiable market influencers, and wherein the multimodal reasoning engine assigns a credibility weight to each identified influencer source based on the source's historical correlation with subsequent market movements, such that the manipulation risk estimate reflects the weighted aggregate behavioral influence of multiple identified sources simultaneously.

24. The system of Claim 1, wherein the explainable alert generation engine generates counterfactual explanations by computing the difference in the manipulation risk estimate between the actual observed signal values and one or more hypothetical alternative signal values, and includes in each delivered alert a statement of the minimum change in observed signals that would have been sufficient to prevent the alert from being generated, thereby enabling users to understand the causal basis of each alert without requiring technical expertise in the underlying model.